YOULAN ZHAO(赵幽兰)

AI for Scientific Systems · Spatiotemporal Modelling · Uncertainty-Aware Inference

uran_uu @outlook.com | +1 631-406-0291

Research Interest

• Modelling complex observational systems using spatiotemporal and statistical learning methods
• Uncertainty quantification and robust inference under noise, missing data, and distribution shift
• Data fusion across heterogeneous sources (sensor, environmental, real-world observational data)
• Structure-aware modelling (temporal dependencies, causal signals, system-level consistency)
• Interpretable and decision-relevant modelling for real-world systems

EDUCATION

Stony Brook University, SUNY – New York, America | **Anhui University** – Anhui, China
Dual B.S. in **Applied Mathematics & Statistics (SBU)** & **Applied Statistics (AHU)**
Expected May 2026

- GPA: 3.58 / 4.00 (Stony Brook University)
- Dean's List, College of Engineering and Applied Sciences (Fall 2025)

- Relevant Coursework:

  Math/Stats: Probability Theory, Applied Stochastic Processes, Time Series Analysis, Multivariate Statistics, Applied Linear Algebra

  CS/AI: Data Structures, Statistical Modeling, Python/R/Matlab Programming, Learning AI Through Visualization (at Columbia University)

  Environmental & Earth Systems: Natural Hazards (environmental systems, risk, and uncertainty)

  Economics: Microeconomics, Macroeconomics, Econometrics

- Honors: MCM Award (2024), Outstanding Volunteer (Shanghai Disabled Persons' Federation, 2020)

Publication (Environmental Systems & Data-Driven Modeling)

1. Wang, X., **Zhao, Y.**, et al. "CAAD: Causality-Aware Multivariate Time Series Anomaly Detection via Multi-Scale Alignment and Structural Causal Consistency." Submitted to KDD 2026 (ADS Track).
   - Proposed a multi-scale Transformer architecture with asymmetric self-supervised alignment (BYOL-style) for robust time-series representation learning
   - Developed gradient-based causal diagnostics (Granger-based) to monitor structural consistency during inference
   - Designed uncertainty-aware anomaly scoring combining residual statistics (median/MAD) with structural deviation

- Demonstrated that causal consistency violations reveal failure modes undetectable by accuracy metrics, critical for trustworthy AI systems

2. Cheng, X., **Zhao, Y.**, Cheng, H., et al. (2024). "Fuzzy logic in the analysis of interspecific competition of wandering spiders feeding on Ectropis obliqua (Lepidoptera: Geometridae) in tea plantations of China." *International Journal of Pest Management*. DOI:10.1080/09670874.2024.2389241

- Designed uncertainty-aware models combining fuzzy logic and grey correlation
- Modeled nonlinear interactions under noisy real-world data
- Focused on robustness under low-quality observations

3. Cheng, X., **Zhao, Y.**, Chen, S., et al. (2025). "Interspecific Competition Between Natural Enemies of Juice-Sucking Pests in Tea Gardens." *Biocontrol Science and Technology,* 35(4), 407–428. DOI:10.1080/09583157.2025.2457056

- Applied multivariate statistical methods to dynamic environmental systems
- Worked with noisy observational datasets, focusing on uncertainty-aware inference

Research & Quantitative Experience

1.Research Assistant - Stony Brook University (Supervisor: Prof. Tengfei Ma) *(June 2025 - Present)*

- Developed Transformer-based models (TST, PatchTST) for time-series reconstruction and anomaly detection
- Designed models robust to noise, missing data, and distribution shift
- Built self-supervised learning pipelines in PyTorch
- Applied models to multi-domain datasets (environmental, financial, biomedical)
- Developed explainable AI tools for validating model behavior in scientific workflows

2.Nonlinear Portfolio Optimization & Empirical Analysis - Anhui University (*2025*)

- Built data-driven optimization models for portfolio allocation
- Applied predictive modeling and uncertainty-aware decision frameworks
- Strengthened understanding of model evaluation under uncertainty

3.Undergraduate Innovation Project (*2024*)

- Designed interval data fusion models (IOWA operator)
- Improved robustness under uncertainty by ~15%
- Relevant to multi-source scientific data fusion

4.Mathematical Contest in Modeling (MCM2023)

- S-Award Winner
- Developed predictive models and optimization strategies
- Produced technical reports using LaTeX

TECHNICAL SKILLS

Machine Learning / AI: Transformers (TST, PatchTST), MAE, self-supervised learning, Uncertainty-aware modeling, anomaly detection, explainable AI

Environmental & Statistical Modeling: Time-series reconstruction, data fusion, fuzzy logic, grey correlation

Programming: PyTorch, Python, R, MATLAB

Languages: Mandarin (Native), English (IELTS 7), Japanese (JLPT N2)

LEADERSHIP & ACTIVITIES

1. Cambridge University | Economy research (2024): Made a few talks about Nobel laureates in Economics.
2. Student Journalist | AHU Media Center (2022--2024): Led interviews and wrote articles for AHU's official WeChat platform (50K+ total views).
3. Founder & Band Leader | Rock Band (2023--now): Organized 4 live performances; managed team collaboration and event logistics.
4. Competitive Tennis Doubles Player | College Tennis Open (2023): Ranked Top 8 in Women's Doubles among 30+ teams.
5. Yangtze Finless Porpoise Program, SEE Foundation (2023–2024): Designed science communication modules; experience with **real-world data uncertainty & bias**